

DSAP:

- Chapter 2:** [6 marks]
- Energy and power signal
 - Check whether the signal is periodic or not. & calculate period
 - Calculate energy of the system
 - Plot the given system
- Imp {
- What is ramp signal?
 - Unit step signal, unit impulse signal, Exponential signal, sinusoidal signal. Defⁿ with example. (Plot)
 - Relationship between unit step, ramp and delta.
- √Imp • Even and odd part of signal.

- Chapter 3:** Linear Time Invariant system: [6 marks]
- Check linearity, time invariance.
 - Define LTI, Convolution.
- √Imp • Convolution Numerical i.e. o/p of LTI system
Mathematical, Graphical, Overlap & Add
- Also check the answer.

- Chapter 10:** Frequency Response [8-10 marks]
- Plot the magnitude and phase spectrum of the given signal.

- Chapter 11:** Realization of Digital Filters [8-12 marks]
- DFI & DFII structure [4 marks]
 - Lattice structure of an FIR filter: all pole [8] / all zero [8] / pole-zero [12 marks]

[6 marks]

Chapter - 8 : Z transform

• Convolution Property

• ROC

• Properties of ROC

3 marks [Sure Question]

• 72 chains

Imp

Find the ROC of given signal.

[3 to 6 marks] Sure Question. ①

• Time Shifting Property

Imp • Inverse Z-transform.

[24 to 32 marks]

Chapter 9: Digital Filter Design

IIR Filter Design (12-20)

1. Conversion of analog filter to digital

a. Impulse Invariance method.

b. Bilinear transformation method.

2. Approximation techniques

a. Butterworth $\checkmark$ - VImp

b. Chebyshev

c. Cauer/Elliptic

d. Bessel Thompson

Theory:

• Gibbs's Oscillation.

• Optimum filter.

• Bilinear transformation

• Why Kaiser is better than others?

3. Special transformation in Z-domain.

FIR Filter Design (3-14)

1. Windowing

a. Window: Rectangular

Bartlett / Triangular

Blackmann

Hamming

Hanning

Kaiser - VImp.

2. Filter design using window

marks [Sure] 3. Remez - Exchange algo. VImp (AKA Optimum Filter / Parks McClellan / Alternation)

• Optimum filter.

NOT
REQUIRED

Chapter 4: Fourier series

- Properties of DFTS
- Spectral Coefficients
- Process of calculating Fourier Series Coefficient

Chapter 5: Fourier Transform

- Discrete-Time Fourier Transform
- Magnitude & Phase Spectrum

Chapter 6: Energy & Power

- Calculate energy & power signal
- Parseval's Theorem
- Defⁿ of Energy & Power Signal

Chapter 7: DFT & FFT

- DFT Defⁿ
- Properties of DFT

2/1 ques.
(1 to 3 marks)

- Comparison of FFT to DFT
- Convolution property of DFT
- Zero Padding
- Need for DFT
- How fast is FFT?

1 ques. → V Imp

1 ques. → V Imp

• Circular Convolution numerical - 4 to 6 marks

• DIT-FFT & DIF-FFT - 6 marks

Chapter 1 & 12: Not that Imp